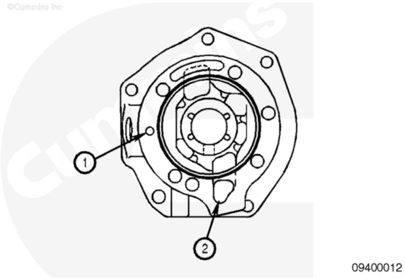
Trainings

General Information

⚠ CAUTION ⚠

Installation of an air compressor on a fuel pump drive housing will result in failure because of a lack of lubrication.



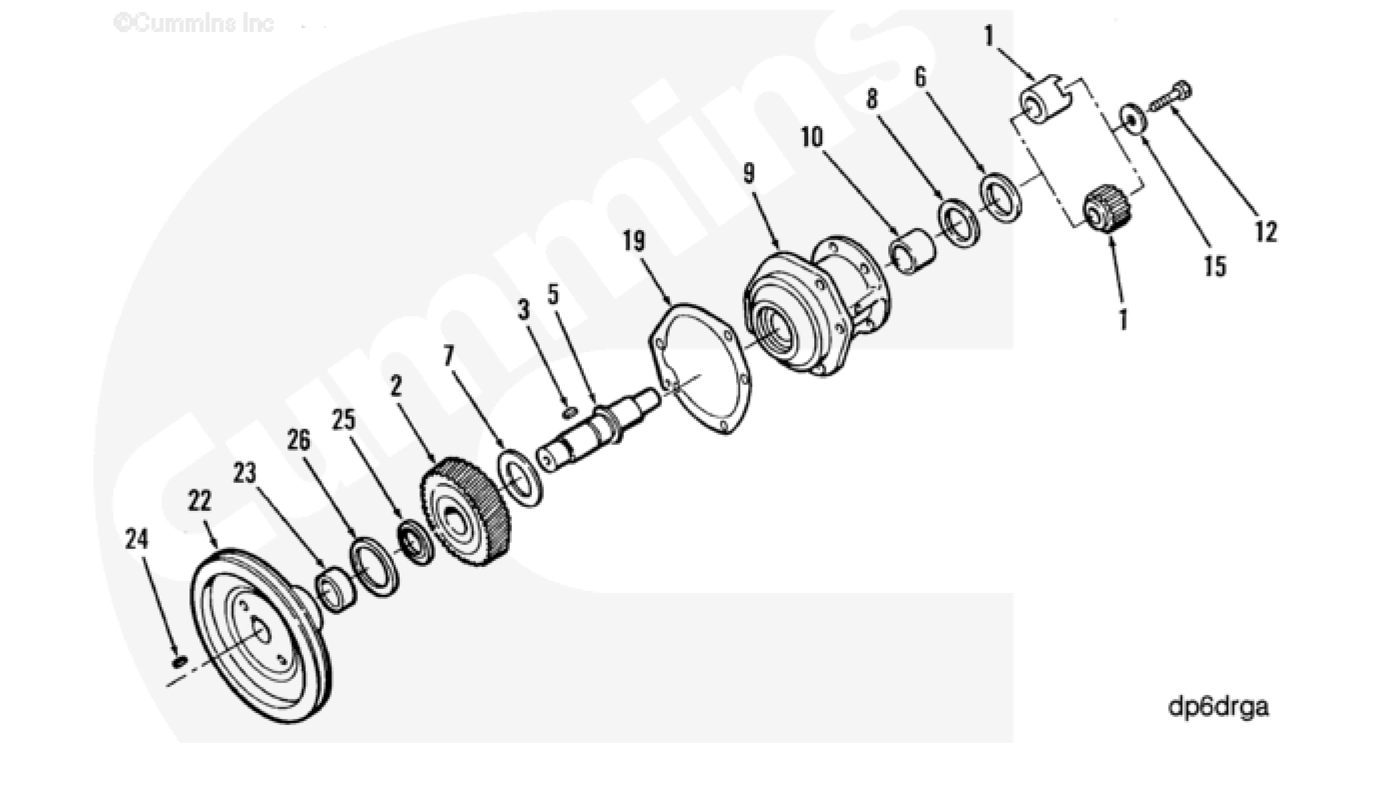
The fuel pump drive housing can be manufactured from aluminum or cast iron. The aluminum housing does **not** contain a bushing, and does **not** require thrust washers.

The housings are available in two designs, with and without provisions for mounting an air compressor. Those without provisions for mounting an air compressor do **not** have provisions for air compressor oil supply (1) and drain back (2). The housing with an air compressor drive has provisions for a splined sleeve type coupling. The procedures are identical for both designs.

The fuel pump/compressor drive gear have stamped marks. The stamped marks **must** be aligned properly with the camshaft idler gear so the valve and injector adjustment marks on the accessory drive pulley are oriented correctly.

The retainer capscrew on the air compressor drive (splined half coupling) is special. It has a drilling in it that provides lubrication of the splined coupling. The retainer capscrew on the fuel pump drive (lovejoy half coupling) does **not** have an oil drilling.

Exploded View



1. Special capscrew
2. Plain washer
3. Love-joy type coupling (upper)
4. Spline type coupling (lower)
5. Clamping washer
6. Thrust bearing
7. Fuel pump drive bushing
8. Fuel pump drive housing
9. Fuel pump support gasket
10. Accessory drive shaft
11. Plain woodruff key
12. Thrust bearings
13. Air compressor and fuel pump drive gear
14. Oil slinger
15. Oil seal
16. Wear sleeve
17. Accessory drive pulley
18. Plain woodruff key

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused dispose of in accordance with local environmental regulations.

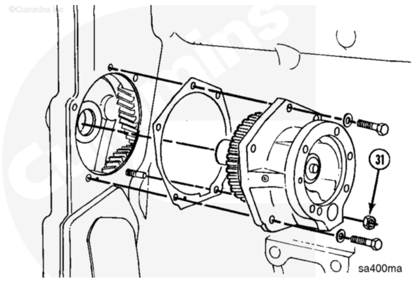
Note : It is not necessary to drain the cooling system if an air compressor not mounted on the engine.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html).
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5 (/qs3/pubsys2/xml/en/procedures/18/18-005-016-tr.html).
- Remove the air compressor. Refer to Procedure 012-014 in Section 12 (/qs3/pubsys2/xml/en/procedures/18/18-012-014-tr.html).
- Remove the accessory drive pulley. Refer to Procedure 009-004 in Section 9 (/qs3/pubsys2/xml/en/procedures/18/18-009-004-tr.html).
- Remove the accessory drive seal. Refer to Procedure 001-003 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-003.html).

Remove

⚠ CAUTION ⚠

The woodruff key must be removed before removing the fuel pump drive assembly. The bushing will be damaged if the woodruff key is not removed.



Remove the woodruff key.

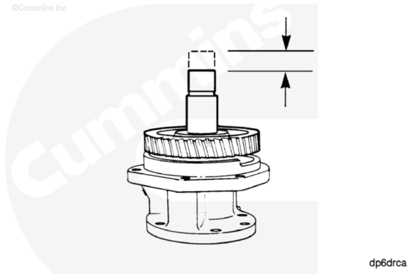
Remove the four capscrews and nut (31).

Remove the fuel pump drive assembly.

Inspect for Reuse

Measure the fuel pump drive end clearance.

Fuel Pump Drive End Clearance		
mm		in
0.05	MIN	0.002
0.30	MAX	0.012

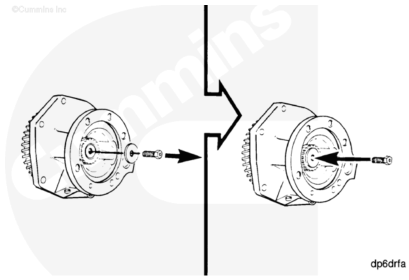


If the fuel pump drive end clearance is **not** within specifications, the fuel pump drive **must** be reconditioned.

Disassemble

⚠ CAUTION ⚠

To reduce the possibility of damage to the shaft, the capscrew must be installed without the washer.



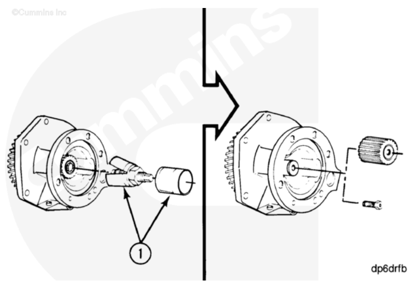
Remove the special capscrew and washer.

Install the capscrew into the shaft.

Remove the splined coupling with coupling puller, Part Number 3376663, or equivalent.

Remove the lovejoy (1) type coupling with a 3-jaw puller.

Remove the capscrew.



Note : Aluminum housings do not contain thrust bearings.

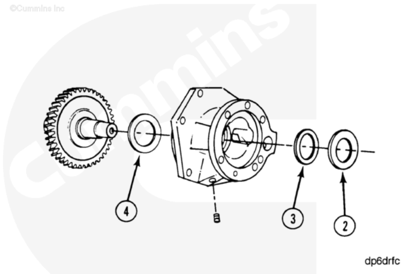
Remove the clamping washer (2).

Remove the inner thrust bearing (3).

Remove the gear and shaft assembly.

Remove the outer thrust bearing (4).

Remove the pipe plug from the housing.

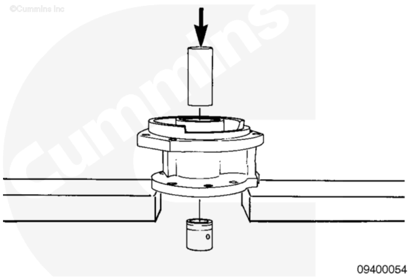


Note : Aluminum housings do not contain a bushing.

Only remove the bushing if it is **not** within specifications or damaged.

Support the housing in an arbor press.

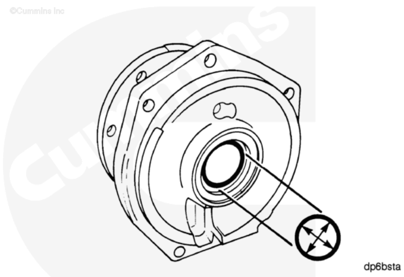
Remove the bushing with a mandrel and an arbor press.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Clean the parts with solvent.

Inspect the parts for damage.

Damaged parts **must** be replaced.

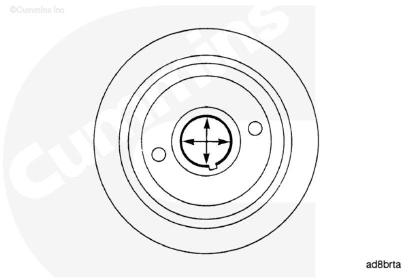
Note : An aluminum housing does not contain a bushing. The inside diameter is identical to the bushing diameter in the cast iron housing.

Measure the inside diameter of the bushing.

Fuel Pump Drive Bushing Inside Diameter		
mm		in
33.43	MIN	1.316
33.50	MAX	1.319

If the bushing is **not** within specifications, it **must** be replaced.
For aluminum housings, the housing **must** be replaced.

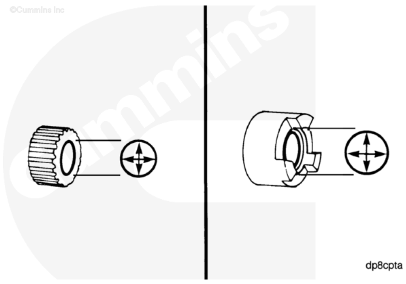
Measure the inside diameter of the pulley.



Pulley Inside Diameter		
mm		in
34.912	MIN	1.375
34.938	MAX	1.376

If the pulley is **not** within specifications, it **must** be replaced.

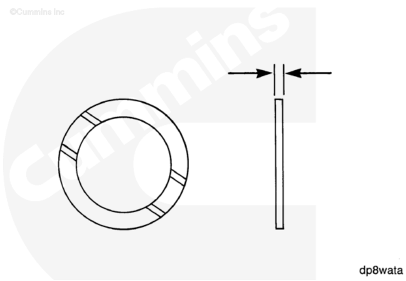
Measure the inside diameter of the coupling.



Lovejoy Coupling Inside Diameter		
mm		in
25.425	MIN	1.001
25.438	MAX	1.002

Spline Coupling Inside Diameter		
mm		in
25.400	MIN	1.000
25.425	MAX	1.001

Check the thrust bearing grooved surface for damage.



The thrust bearing **must** be replaced if the surface is damaged.

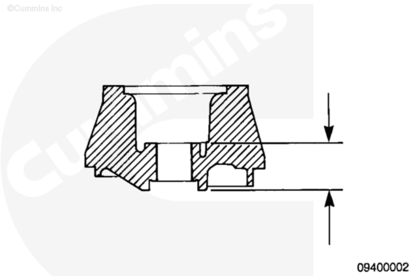
Measure the thickness of the thrust bearing.

Thrust Bearing Thickness		
mm		in
2.36	MIN	0.093
2.41	MAX	0.095

If the thrust bearing is **not** within specifications, it **must** be replaced.

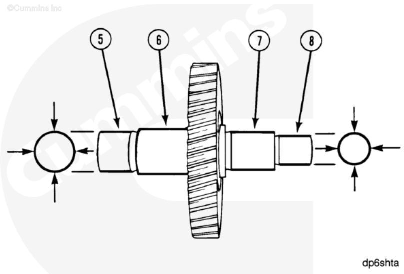
On aluminum housings **only** check the two machined surfaces for damage and measure the depth.

Aluminum Housing Machined Surface Depth		
mm		in
45.54	MIN	1.793
45.67	MAX	1.798



If the housing is damaged or **not** within specifications, it **must** be replaced.

Check the gear teeth for damage.



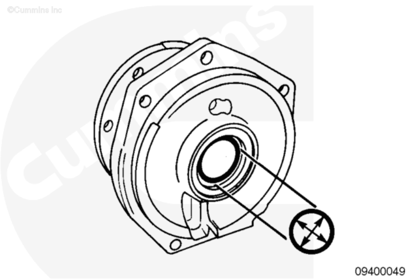
If the gear teeth are damaged, the gear **must** be replaced. Refer to Procedure 009-013 (/qs3/pubsys2/xml/en/procedures/18/18-009-013.html) in Section 9.

Measure the shaft outside diameter.

Shaft Outside Diameter			
	mm		in
(5)	34.963	MIN	1.3765
	34.976	MAX	1.3770
(6)	34.662	MIN	1.5616
	39.674	MAX	1.5620
(7)	33.300	MIN	1.3000
	33.330	MAX	1.3120
(8)	25.476	MIN	1.0030
	25.489	MAX	1.0035

If the shaft is **not** within specifications, it **must** be replaced.

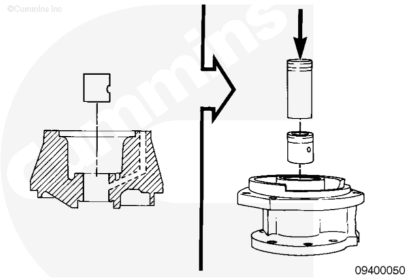
Measure the inside diameter of the cast iron housing bushing bore.



Bushing Bore Inside Diameter without Bushing		
mm		in
36.73	MIN	1.446
36.75	MAX	1.447

If the bushing bore is **not** within specifications, the housing **must** be replaced.

Assemble

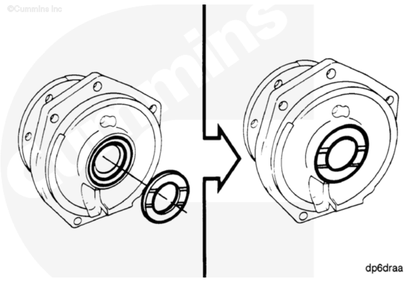


Support the housing in an arbor press.

Align the oil hole in the bushing with the oil drilling in the housing.

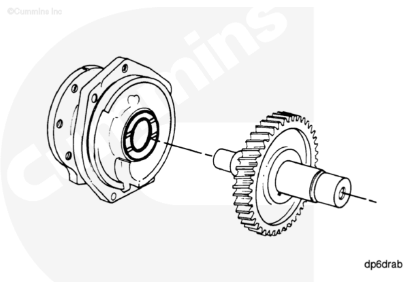
Press the bushing with a mandrel and the arbor press.

Both ends of the bushing **must** be positioned even with, or below, the surface of the housing.



Note : Aluminum housings do not contain thrust bearings.

Position the grooved surface of the thrust bearing on the housing as illustrated in the graphic.



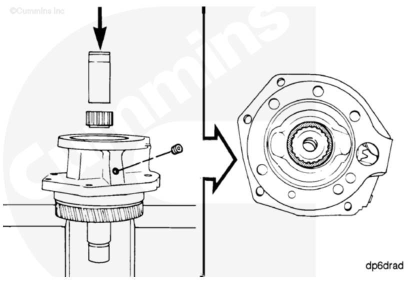
Lubricate the grooved surface of the thrust bearing with Lubriplate®, Number 105, Part Number 3163086, or equivalent.

For aluminum housings, lubricate the machined thrust surface.

Install the gear and shaft assembly into the shaft.

With the grooved surface positioned up, slide the thrust bearing over the shaft.

Install the clamping washer with the beveled edge pointing down toward the thrust bearing.



Support the gear or the shaft in an arbor press.

Use a mandrel and arbor press to press the coupling on the shaft until it touches the clamping washer.

The clamping washer **must** be positioned tightly between the coupling and the shoulder of the shaft.

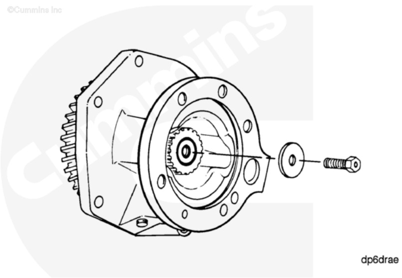
Install the pipe plug into the housing.

Tighten the pipe plug.

Torque Value: 8 n•m [75 in-lb]

⚠ CAUTION ⚠

To reduce the possibility of engine damage the coupling capscrew must contain an oil drilling if an air compressor is to be mounted on the engine.



There are two diameters of coupling capscrews. The capscrew torque depends on the diameter.

Install the washer and capscrew.

Torque Value:
9.5 mm [3/8 in]

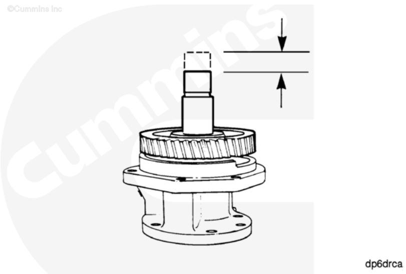
1. 45 n•m [33 ft-lb]

Torque Value:
12.7 mm [½ in]

1. 100 n•m [75 ft-lb]

Rotate the shaft to verify correct assembly.

Measure the end clearance with a dial indicator.



Fuel Pump Drive End Clearance		
mm		in
0.05	MIN	0.002
0.30	MAX	0.012

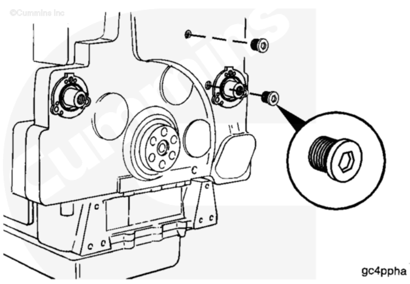
If the end clearance is **not** within specifications, make sure the coupling is positioned tightly against the clamping washer.

Oversize thrust bearings are available to adjust the end clearance.

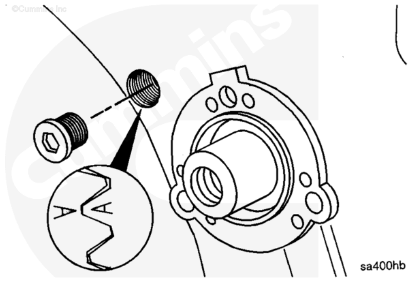
Install

Remove the two straight threaded o-ring plugs from the timing holes in the front cover.

Check the index mark alignment.



Do not use the “A” on the camshaft idler gear for the accessory drive alignment unless the “X” marks on the camshaft and the camshaft idler gears are aligned and centered in the upper timing plug hole. If the “X” marks are not visible in the upper plug hole, rotate the engine until the “X” marks on the camshaft gear and the camshaft idler gear are centered in the upper timing plug hole.



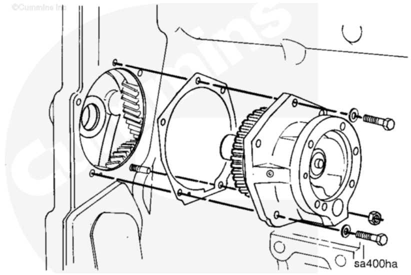
Install the fuel pump drive gasket.

Install the fuel pump drive so the “A” on the fuel pump drive gear is centered in the lower timing plug.

Install the capscrews and nut.

Tighten the capscrews and nut.

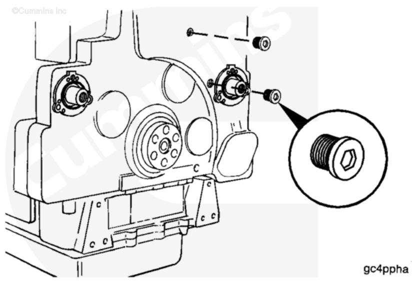
Torque Value: 45 n•m [33 ft-lb]



Install the straight threaded o-ring plugs.

Tighten the o-ring plugs.

Torque Value: 25 n•m [20 ft-lb]



Finishing Steps

- Install the accessory drive seal. Refer to Procedure 001-003 in Section 1 (/qs3/pubsys2/xml/en/procedures/18/18-001-003.html).
- Install the accessory drive pulley. Refer to Procedure 009-004 in Section 9 (/qs3/pubsys2/xml/en/procedures/18/18-009-004-tr.html).
- Install the air compressor. Refer to Procedure 012-014 in Section 12 (/qs3/pubsys2/xml/en/procedures/18/18-012-014-tr.html).
- Install the fuel pump. Refer to Procedure 005-016 in Section 5 (/qs3/pubsys2/xml/en/procedures/18/18-005-016-tr.html).

- Fill the cooling system. Refer to Procedure 008-018 in Section 8 (/qs3/pubsys2/xml/en/procedures/18/18-008-018-tr.html).
- Operate the engine to 70°C [160°F]. Check for leaks.